## ANALYSTLAB AFRICA

**Data Science Internship Program**

**Data Preprocessing Report: Heart Failure Prediction**

### Business Problem

Heart failure occurs when the heart cannot pump enough blood to meet the body's needs. It's usually triggered by other health conditions affecting the heart muscles, including coronary artery disease, heart attack, high blood pressure, heart valve disorders, congenital defects, cardiomyopathy, myocarditis, diabetes, and pulmonary hypertension.

While heart failure is often caused by underlying health conditions, each individual's health risks are unique. Many factors like age, family history, and lifestyle can contribute, but early detection with advanced diagnostic tools can help manage and reduce the likelihood of serious damage.

Therefore, a heart failure prediction model would be effective in detecting the likelihood or the presence of heart failure in individuals.

### Preprocessing Decisions

- Cholesterol initially showed no null values. Upon confirmation of the number of zeroes, there was a suspicion that they could be null values written as 0. To resolve, the rows with the 0 were converted to NaN (null values), and it was replaced with the median of the column. The median was selected to replace the non-null values as it is not mostly affected by extreme values or outliers.
- The same logic was applied to the RestingBP column.

### Feature Engineering

- A new column ' Age group' was created from the age column. This is because age groups are more easily analyzed than raw ages.
- Every column was important and clinically relevant in predicting heart failure.

### Feature Encoding

#### Label encoding

Binary categorical variables like **'Sex'** and **'ExerciseAngina'** are binary with no ordering (i.e M/F, Yes/No), so Label encoding is sufficient as it avoids dimensionality.

#### One-hot encoding

One-Hot encoding was used for categorical variables without a particular order. It avoids implying a false ordinal relationship that Label Encoding would introduce.

### Feature Scaling

StandardScaler is appropriate for this project because it involves numeric columns like Age, RestingBP, Cholesterol, MaxHR, and OldPeak, and placing these features on the same scale by giving them a mean of 0 and a standard deviation of 1.